Journal of South American Earth Sciences 126 (2023) 104327



Contents lists available at ScienceDirect

Journal of South American Earth Sciences

journal homepage: www.elsevier.com/locate/jsames





New magnetostratigraphy constrains in the southern Paraná magmatic province (Herveiras and Gramado Xavier areas), Rio Grande do Sul state, South Brazil: Implications for the timing between volcanic sources

M. Irene B. Raposo [a,*], Thales Pescarini [a], Letícia F. Guimarães [b], Melina C.B. Esteves [a]

[a] Instituto de Geociências, Universidade de São Paulo, Rua do Lago 562, 05508-80, São Paulo, Brazil
[b] Instituto de Geociências, Universidade Federal da Bahia, R. Barão de Jeremoabo, s/n, 40170-290, Salvador, Brazil

ARTICLE INFO

Keywords:
Paleomagnetism
Silicic rocks
Basalt
Paraná magmatic province
Paraná-etendeka

ABSTRACT

This paper undertakes a new paleomagnetic study of 51 lava flows that crop out in Southern Paraná Magmatic Province. The volcanic rocks are basalt of low-Ti content type (18 flows) and Palmas silicic type (33 flows) corresponding to the Caxias do Sul, Barros Cassal, and Santa Maria sub-types covering all lithostratigraphic units of the region. This study is part of a broad magnetic study in the Brazilian part of the Paraná-Etendeka LIP to understand the temporal relationship between the volcanic units of different compositions allowing the investigation of the contemporary nature of the multiple sources. The characteristic remanent magnetization (ChRM) for the lavas was obtained by alternating field (a.f.) and thermal demagnetization methods, and is interpreted as a thermo-remanent magnetization carried by stable pseudo-single domain magnetite grains, being acquired during the cooling of the lava flows. The high-quality of the ChRM results from both areas allows us to interpret that the emplacement of all volcanic rocks was geologically very fast, as verified by the symmetrical direction of both polarities (normal and reverse i.e. positive reversal test) of the geomagnetic field associated with the stratigraphy of all lava flows. The passage from pahoehoe to rubbly basalts was marked by a change of polarity of the geomagnetic field. Our results suggest that all volcanic rocks were emplaced by at least two volcanic events as shown by the polarity changes, and the sources of basaltic and silicic rocks of different compositions were active at the same time.

1. Introduction

The Cretaceous period in South America was marked by the occurrence of a voluminous magmatic event that preceded the opening of the South Atlantic Ocean, which has culminated in the formation of the Paraná-Etendeka Magmatic Province (PEMP, in Brazil known as Paraná Magmatic Province, PMP), the second largest continental igneous province on Earth. About 2/3 of its estimated preserved volume occur in South America, with around 450,000 km^3 of volcanic and 112,000 km^3 of sub-volcanic rocks extending through the central-west, southeast and south of Brazil, southeastern Paraguay, northern Argentina and eastern Uruguay (Frank et al., 2009, Fig. 1). The Paraná Magmatic Province (PMP) is composed of a broad group of basic, intermediate (andesites) to silicic volcanic rocks that overlies the aeolian sandstones from Botucatu Formation of the Paraná Basin. Tholeiitic basalts and intermediate rocks compose about 97% (vol.) of these rocks, versus only 3% (vol.) of the silicic rocks (dacites and rhyolites). Many classifications are made for these rocks, mainly based on geochemical criteria firstly proposed by Bellieni et al. (1984) and Mantovani et al. (1985), and by other researchers (see below). Significant chemical differences regarding Ti and incompatible elements contents characterize both basalts and silicic units. The tholeiitic basaltic rocks were divided into two main groups high-Ti (>2 wt %), richer in incompatible elements such as Ba, La, Zr, and Sr and poorer in Rb, Th and U, compared to low-Ti basalts (Piccirillo and Melfi, 1988). The silicic rocks, based on geochemical and petrographic characteristics, were divided into Palmas (low-Ti) and Chapecó (high-Ti) types. The Palmas type rocks are, in general, aphyric to microporphyritic dacites and rhyolites that occur associated with the low-Ti basalts (Bellieni et al., 1984; Piccirillo and Melfi, 1988). The Chapecó type rocks are porphyritic trachydacites and occur associated

* Corresponding author.
E-mail addresses: irene@usp.br (M.I.B. Raposo), thales.pescarini@usp.br (T. Pescarini), guimaraesleticia@ufba.br (L.F. Guimarães), melina.cbe@usp.br (M.C.B. Esteves).

https://doi.org/10.1016/j.jsames.2023.104327
Received 13 February 2023; Received in revised form 30 March 2023; Accepted 31 March 2023
Available online 6 April 2023
0895-9811/© 2023 Elsevier Ltd. All rights reserved.

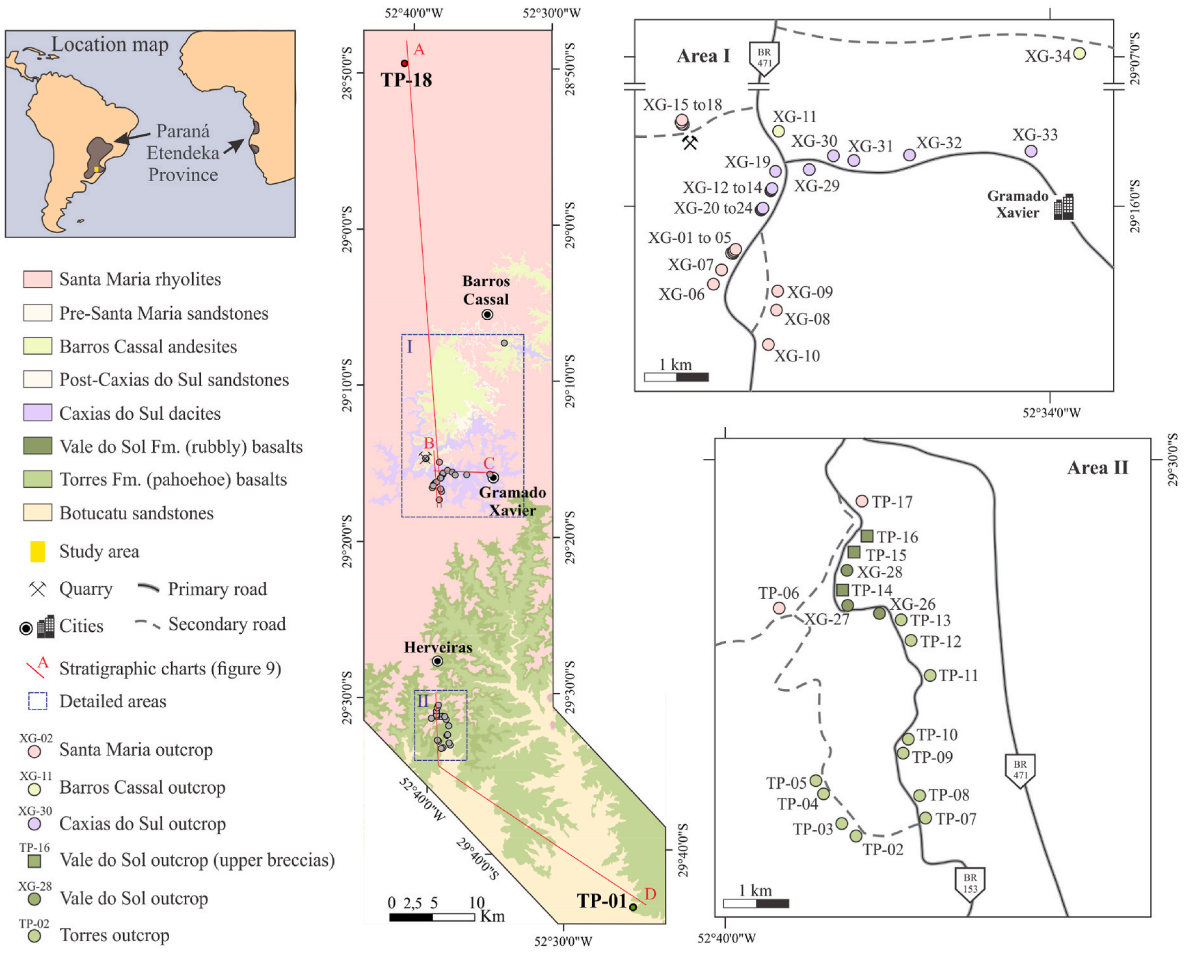


Fig. 1. General geological map showing the two analyzed region that was divided into area #I and area #II, and the location of the sampled sites. The profiles A, B, C and D are presented in Fig. 2.

with the high-Ti basalts (Bellieni et al., 1984; Piccirillo and Melfi, 1988). The low-Ti basalts and silicic Palmas rocks are mainly found in the southern portion of the Province and the high-Ti basalts and silicic Chapecó type occur at northern portion. However, between these two areas, the so-called Central Paraná Magmatic Province, all magma types are found (Piccirillo and Melfi, 1988).

The tholeiitic magmas were further subdivided into six basalt magma types based on their chemical composition (Peate et al., 1990, 1992), the high-Ti basalts were divided into: Pitanga, Paranapanema, Ribeira and Urubici types, whereas the low-Ti basalts were divided into Gramado, and Esmeralda types. On the other hand, recently seeking a formal lithostratigraphy in accordance with the International Stratigraphic Code, Rossetti et al. (2018) proposed the subdivision of the Gramado magma type into, from bottom to top, Torres Formation (more primitive, i.e higher MgO and lower SiO_2 contents, pahoehoe basaltic flows), and Vale do Sol Formation (more evolved sheet-like rubbly basaltic-andesite flows). Based on petrographic and geochemical criteria the Palmas rocks were also subdivided into Caxias do Sul, Anita Garibaldi, Jacuí, Barros Cassal, Clevelândia and Santa Maria subtypes (Peate et al., 1992; Garland et al., 1995; Nardy et al., 2008; Polo and Janasi, 2014). The Chapecó type was subdivided into Ourinhos, Guarapuava and Tamarana subtypes (Peate et al., 1992; Nardy et al., 2008).

In this paper we present new paleomagnetic data of the low-Ti basalts (Torres and Vale do Sol Formations) and related Palmas-type silicic rocks (Caxias do Sul, Barros Cassal and Santa Maria subtypes, Figs. 1 and 2) from the vicinities of Herveiras, Gramado Xavier and Barros Cassal cities (Southern of PMP, Figs. 1 and 2). The main goal of this study is to investigate whether the sources of those units were active simultaneously, especially between the sources of basic and silicic rocks, and verify the behavior of geomagnetic field in the transition between pahoehoe and rubbly morphology of the basic lava flows. The studied area was divided into area #I and #II (Fig. 1), in order to clarify our study. The lava flows in both studied areas are in horizontal position.

2. Geological setting

Isotopic studies and geochemical modeling suggest that the basaltic melts of the PMP magmatism may have derived from mixing between metasomatized asthenospheric mantle and enriched mantle components (Rocha-Júnior et al., 2012), possibly with subsequent crustal contamination (Barreto et al., 2016). On the other hand, an origin related to fractional crystallization of these basalts together with crustal assimilation is proposed for the silicic magmatism (Garland et al., 1995; Guimarães, 2014). The eruptive style of the silicic rocks is still a topic of debate. Although some authors consider the whole silicic volcanism resulting from large-scale explosive events from a central conduit located in the African portion of PEMP (Bryan et al., 2010; Ewart et al., 1998). Recent studies in the Brazilian portion of the Province have suggested local fissural feeder systems, with the predominance of effusive deposits over restricted block-and-ash flows and rheoignimbrites (e.g. Lima et al., 2012: Guimarães et al., 2018a,b, 2019; Luchetti et al., 2018a, b, Polo et al., 2018a, b; Simões et al., 2018). In the studied area, at the vicinities of the cities of Herveiras (to the south) and Barros Cassal (to the north), some authors, based on detailed mapping, geothermobarometry, rheological studies and anisotropy of magnetic susceptibility, defined that the silicic volcanism of the region was

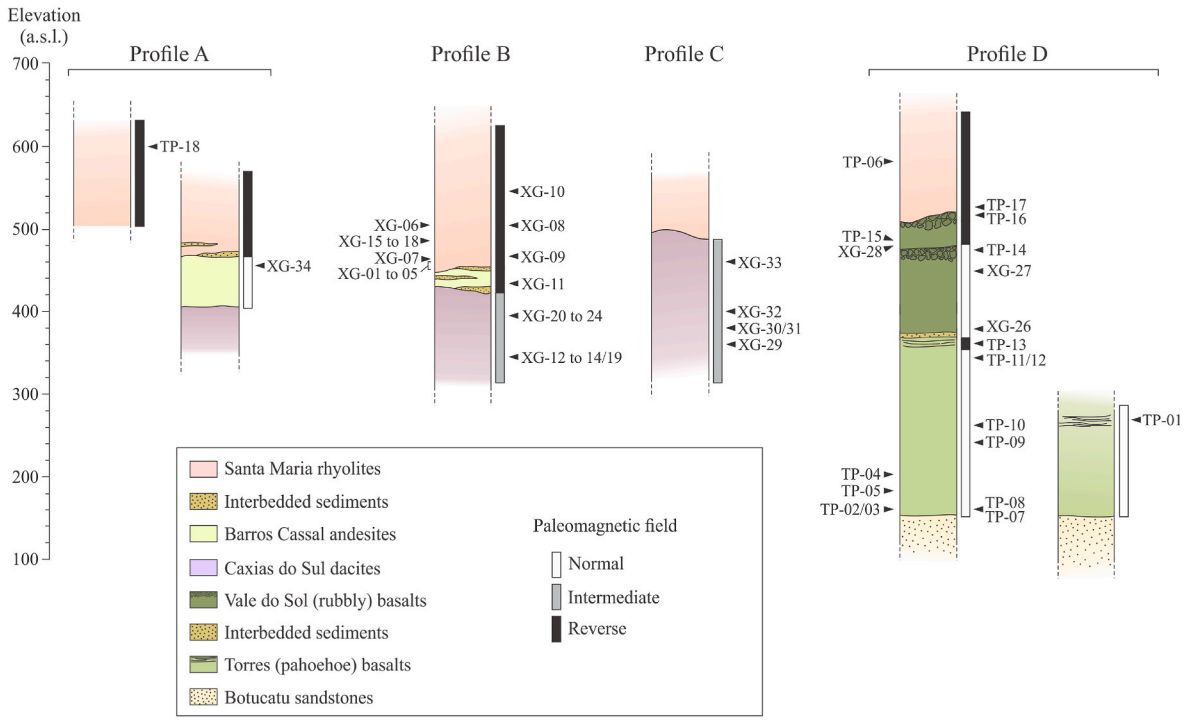


Fig. 2. Stratigraphic columns, modified from Polo and Janasi (2014), Barreto et al. (2014) and Polo et al. (2018). Sample sites'stratigraphic positions are indicated as well as the magnetic polarity. See Fig. 1 for location.

characterized by extrusion of lava flows and lava-domes (Guimarães et al., 2018b; Polo et al., 2018a; b, Giordano et al., 2021).

The time span of Paraná volcanism given by paleomagnetic data from silicic Palmas type and low-Ti tholeiitic basalt from south of PMP rocks is less than 1 Ma (Ernesto and Pacca, 1988), which was confirmed by Ar/Ar ages (Renne et al., 1992). A review of all published ages from the PMP (Baksi, 2018) show that ages obtained by Ar/Ar and U–Pb techniques range between 135 and 134 Ma, with the time duration of the volcanism ~1 Ma. On the other hand, high-precision zircon U–Pb ages from the silicic Palmas (Santa Maria rhyolites and Caxias do Sul dacites) sequences from southern part of PMP showed that rocks were erupted around 133.6 Ma within a narrow time interval of 0.12 ± 0.11 ky (Rocha et al., 2020).

2.1. Area #I – Gramado Xavier region

The stratigraphy of area #I (Fig. 1 and 2) consists of a succession of Palmas type, from bottom to top, dacites (Caxias do Sul subtypes) andesites (Barros Cassal subtype) and rhyolites (Santa Maria subtype). The Caxias do Sul dacites occur as lava domes, lavas lobes (XG-12 to XG-14; XG-19; XG-20 to XG-22; XG-23 and 23XG-24, Fig. 1), and sheet flows (XG-29 to XG-33). The Barros Cassal andesites occur as thin sheet flows (XG-11 and XG-34, Fig. 1), and Santa Maria rhyolite is extensive massive to banded tabular deposits (XG-1 to XG-10; XG-15 to XG-18, Fig. 1, Polo et al., 2018; Guimarães et al., 2018b). Intercalation of sedimentary rocks between the lavas is frequent and constitute important stratigraphic markers.

The lower Caxias do Sul sequence (Fig. 1 and 2, area #I) comprises dacites which are mainly exposed along valleys. These rocks have already been studied by Guimarães et al. (2018b) who shows that they are composed of plagioclase, pigeonite, augite and Ti-magnetite in a glassy to devitrified groundmass. The thin sheet lavas of the Barros Cassal sequence are aphanitic to phaneritic hypocrystalline with augite, labradorite-andesine and Ti-magnetites microlites dispersed in a glassy groundmass (Polo et al., 2018).

The Santa Maria rhyolites occur as extensive plateaus that can exceed 150 m thickness. A 2–5 m obsidian layer locally occurs at the bottom of the sequence while the upper part is dominated by massive to banded (Guimarães et al., 2018a, b). These rocks have low crystal contents comprising microphenocrysts and microlites of plagioclase, pigeonite and Ti-magnetite dispersed in a glassy to devitrified groundmass (Guimarães et al., 2018a, b).

2.2. Area #II – Herveiras region

The stratigraphy of area #II (Figs. 1 and 2) consists of a succession of, from bottom to top, pahoehoe basalts (Torres Formation), rubbly basalts (Vale do Sol Formation), and extensive banded rhyolites (Santa Maria subtype).

The Torres Formation (Figs. 1 and 2) comprises simple and compound pahoehoe lavas, which form individual lobes ranging between 0.5 and 2.5 m thick, (Barreto et al., 2014; Rossetti et al., 2018). These rocks are aphanitic to porphyritic (plagioclase <5% modal volume), have as essential mineralogy plagioclase, augite and Fe–Ti oxides (Ti-magnetite and magnetite) ± olivine, in intergranular and more rarely ophitic to sub-ophitic textures (Barreto et al., 2014).

The Vale do Sol Formation (Figs. 1 and 2) comprises thick basaltic to andesi-basaltic sheet-like lava flows with rubbly morphology and average thickness of 35–45 m (Barreto et al., 2014; Rossetti et al., 2018). The contact between the Torres and Vale do Sol Formations is usually marked by an abrupt non-erosional surface, often marked by the presence of a thin layer of sandstones and by a reversal polarity of the geomagnetic field, as observed by Pescarini and Raposo (2017). The degree of crystallinity ranges from holocrystalline to holohyaline, with a vitreous to cryptocrystalline matrix composed of plagioclase, augite, pigeonite and Ti-magnetite (Barreto et al., 2014). The uppermost portions of the Vale do Sol flows are characterized by a brecciated zone (Figs. 1 and 2) with sub-rounded to sub-angular fragments (~5 and 30 cm) of reddish oxidized basalts. The interstitial material consists of weathered basalt.

Extensive banded rhyolites (Santa Maria subtype), with tabular morphology, overlie the basaltic sequence. In the Herveiras area Santa

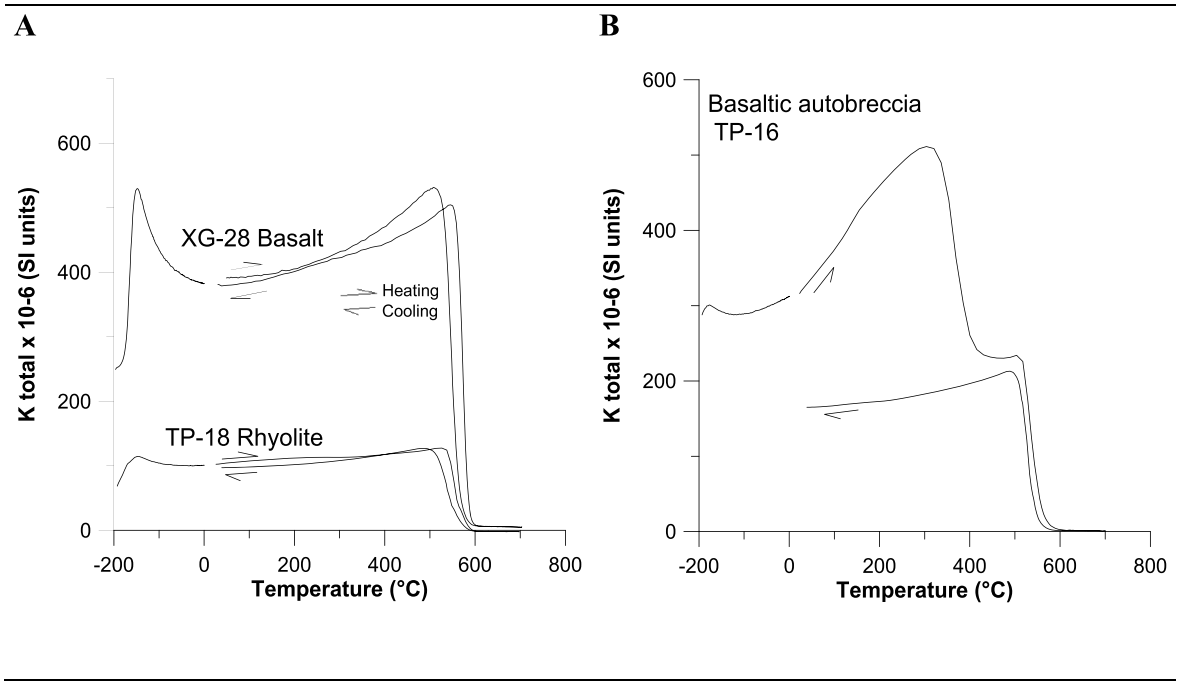


Fig. 3. Representative k-T curves (susceptibility versus low and high temperatures), obtained in argon atmosphere. ***A*** – for basalt from Vale do Sol Formation and rhyolite Santa Maria subtype. ***B*** – for a basaltic autobreccia.

Maria rhyolites occur directly over basalts. These rocks are the same as described in Area #I.

3. Sampling and methods

For the magnetic measurements all sites were sampled with a portable gasoline powered drill, cooled with water. At least 10 cylinders were taken at each drilled site and oriented in-situ with magnetic and solar compass, whenever possible, as well as a clinometer. In general, sampling was performed at fresh road cuts where good exposures are available. A total of 30 sites from Area #I (14 at Caxias do Sul dacites, 14 at Santa Maria rhyolites and 2 at Barros Cassal subtypes, Figs. 1 and 2), and 21 sites from Area #II (12 at Torres Formation basalt, 3 at Vale do Sol Formation basalts, 3 at upper basaltic autobreccias and 3 at Santa Maria rhyolites, Figs. 1 and 2). For the measurements the cylinders were cut into specimens of ~2.2 cm height and 2.5 cm in diameter. At least 3 specimens were cut from each sampled cores.

A detailed rock magnetism study was performed through rock-magnetic experiments such as continuous low-field susceptibility versus low and high temperature curves (k-T curves, Hrouda, 1994); partial anhysteretic remanent magnetization acquisition curves (pARM); isothermal remanent magnetization (IRM) acquisition curves, hysteresis loops and FORC measurements, to decipher and investigate the magnetic carriers of the remanent magnetization. All magnetic measurements were performed at the Laboratory of Magnetic Anisotropies and Rock Magnetism of the Geosciences Institute of São Paulo University, except FORC diagrams that were obtained at the Laboratory of Paleomagnetism and Rock Magnetism (USPMag) of the Institute of Astronomy, Geophysics and Atmospheric Sciences of São Paulo University.

3.1. Rock magnetism properties

The k-T curves from low-temperature (from about −195 °C to room temperature) susceptibility were recorded using a CS3-L apparatus coupled to a Kappabridge (KLY-4S) instrument (Agico Czech Republic). The high K-T curves were carried out in an Ar atmosphere using a CS-3 apparatus coupled to the KLY-4S. Specimens were progressively heated up to 700 °C and subsequently cooled to room temperature. These experiments were performed on one sample from each site.

The acquisition of partial artificial anhysteretic remanent (pARM) values consisted of applying a steady field (DC field) between two chosen values (a.f. Window, H_1 and H_2, $H_1 < H_2$) of a decaying alternate field (a.f.) peak (H), while the rest of the assemblage is demagnetized from a peak field $H > H_2$ (Jackson et al., 1989). The specimens were exposed to an a.f. peak of 95 mT and DC field of 0.16 mT with an a.f. Window width of 10 mT. a.f. demagnetization at 120 mT (through DT-2000 tumbler from ASC Scientific) was applied after each pARM acquisition. The remanence was measured on JR5A (Agico Czech Republic). The pARM was imposed on the specimens using a Molspin (Molspin, Newcastle-upon-Tyne, UK) alternating–field demagnetizer as the source of the alternating magnetic field, with superimposition of a steady field (DC field) attained by a (home-made) small coil inside and coaxial to the demagnetizer, which was controlled by a Molspin apparatus.

After the last a.f. demagnetization steps the specimens in question were subjected to an IRM acquisition curves protocol in progressively increasing magnetizing fields using a impulse magnetometer (MMPM9, Magnetic Measurements). The experiment was performed by stepwise fields from 5 mT up to 600 mT or 2T steps of 20 or 40 mT.

Hysteresis measurements at room temperature were performed using a vibrating sample magnetometer (VSM-Nuvo, Molspin, Newcastle-upon-Tyne, UK) in fields up to 1 T.

FORC measurements were obtained in a Vibrating Sample Magnetometer (Princeton Measurements Corporation MicroMag, VSM Model 2900) with the parameters: (i) $Bu_{min} = -100$ mT and $Bu_{max} = 100$ mT; $Bc_{min} = 0$ mT and $Bc_{max} = 150$ mT; (ii) averaging time = 0.5s; (iii) saturation field = 300 mT; (iv) pauses of 1s at the saturation, calibration, and reversal fields; (v) number of FORCs (NFORCs) = 170–200; and (vi) continuous field sweep mode. The FORCinel software package (Harrison and Feinberg, 2008) was used to process the FORCs. The smoothing factors (SF) used during the processing of FORCs varied between 4 and 7.

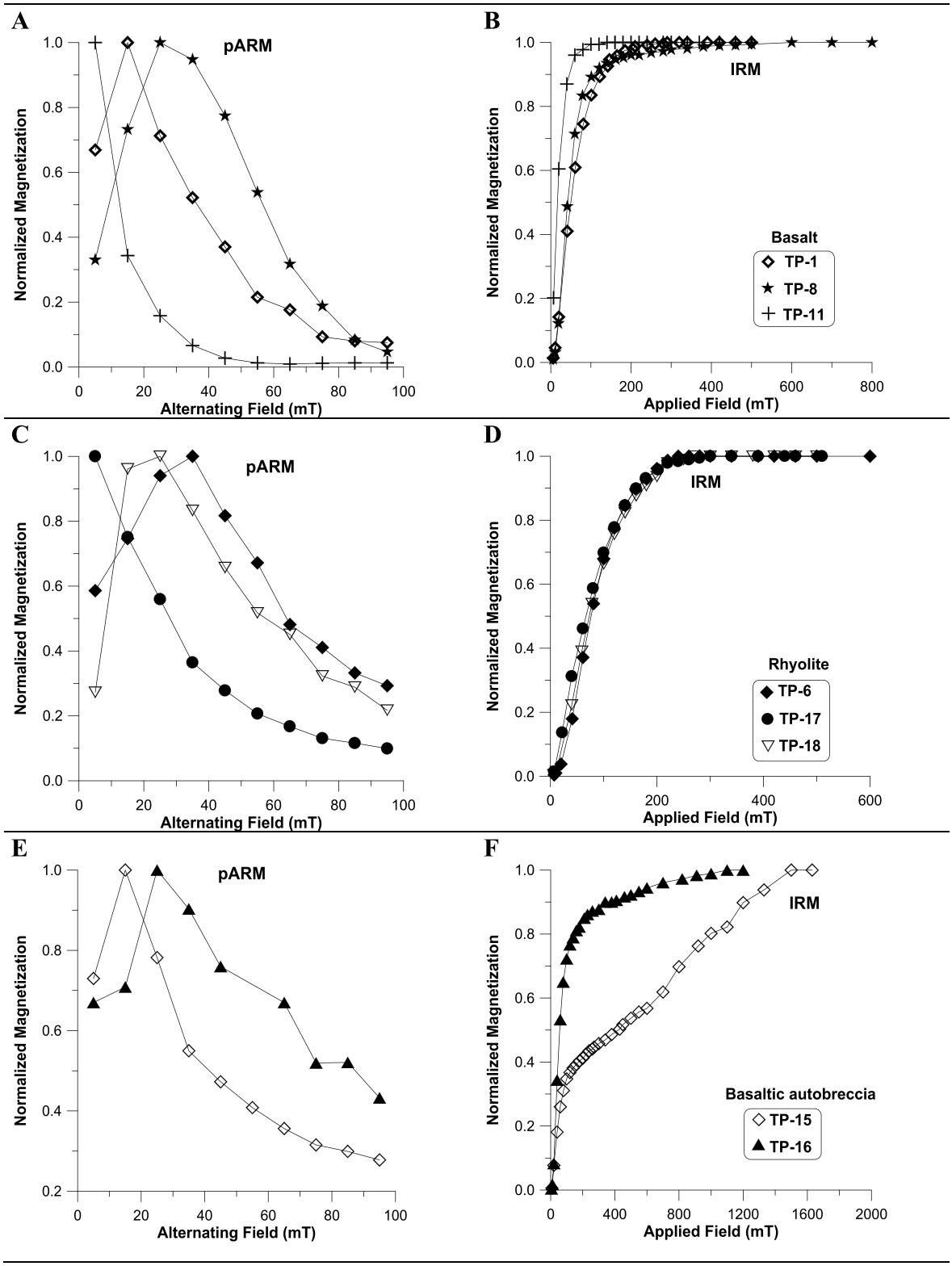


Fig. 4. Examples of rock magnetism for specimens from basalts (Torres and Vale do Sol Formations.), Santa Maria rhyolites and basaltic aubreccias. ***A-C-E-*** Remanent coercivity spectra derived from artificial partial anhysteretic remanence acquisition (see text for detail), remanence intensities are normalized to the highest value of partial remanence. ***B–C-E-*** Isothermal remanence magnetization (IRM) acquisition curves, intensities of remanences are normalized to saturation of IRM (SIRM) versus field strength.

3.2. Paleomagnetism – identification of the magnetization components

To identify the components of the natural remanent magnetization (NRM), samples were subjected to both alternating field (a.f.) and thermal stepwise demagnetizations. One selected specimen from each site was submitted to a detailed stepwise a.f. demagnetization (pilot) up to 100 mT or 150 mT in steps of 5 mT up to 100 mT and in steps of 10 mT up to 150 mT using a Molspin tumbler (Molspin, Newcastle-upon-Tyne,

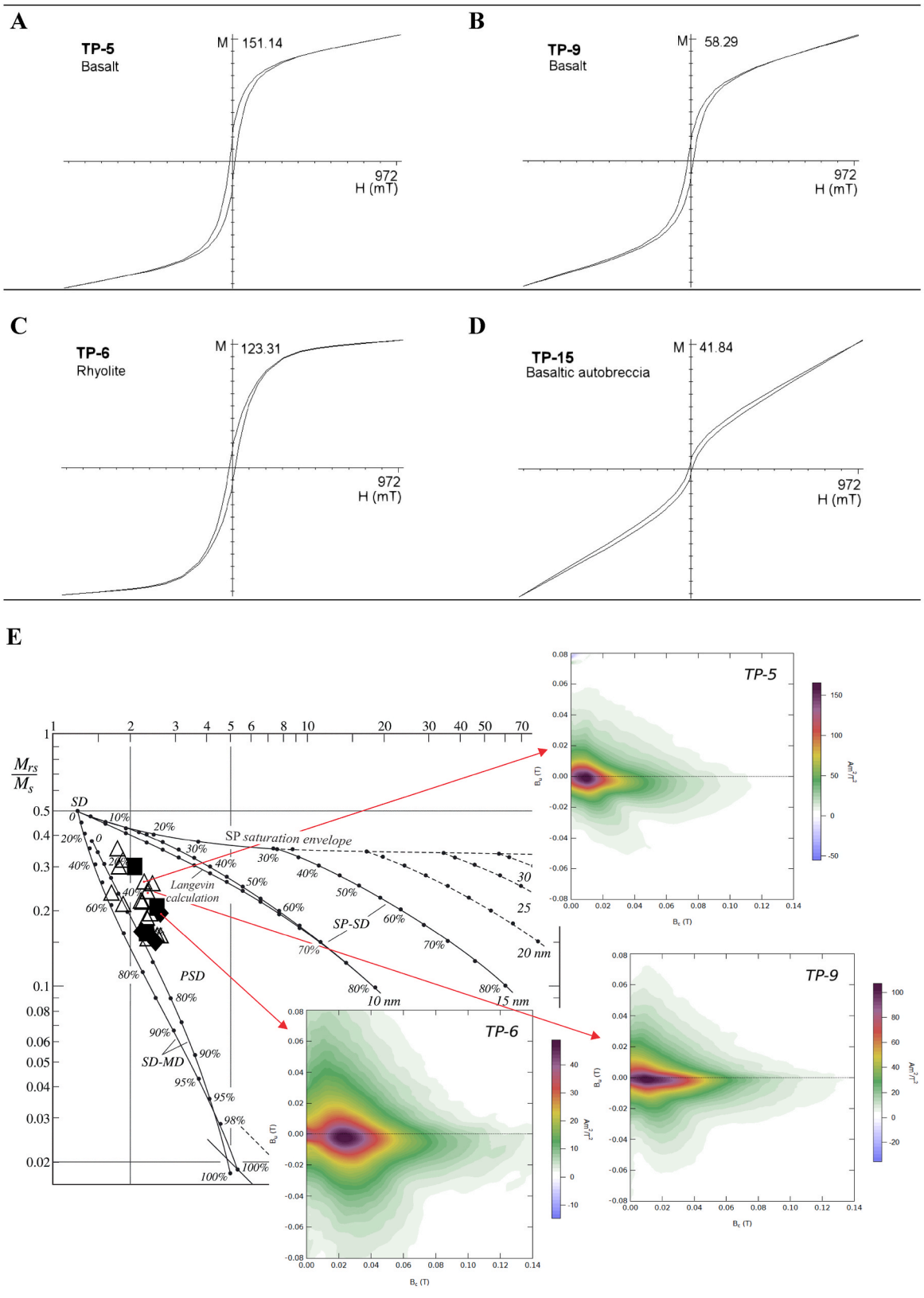


Fig. 5. Hysteresis loops for basalts (Torres Formatiom; ***A and B***), Santa Maria rhyolites (***C***) and basaltic autobreccias (***D***). M is magnetization in μAm^2 and H is applied field. Loops are not corrected for paramagnetic slopes. ***E-*** Dunlop's plot (2002); Hcr/Hc *vs* Mrs/Ms (hysteresis parameters) Hcr is remanent coercivity; Hc is coercive force; Mrs is remanent saturation magnetization; Ms is saturation magnetization; SD is single domain, PSD is pseudo-single domain and MD is multi domain. Open triangles = basalts, full diamonds = rhyolites, full squares = basaltic autobreccias. Inset FORC diagrams.

Table 1
Paleomagnetic results for the volcanic rocks from area #I.

Site	Lat.	Long.	El.	N/n	Mean magnetization direction					Virtual Palemagnetic Pole			
					Dec.	Inc.	α_{95}	k	Pol	PLat.	PLong.	dm	dp
	(°S)	(°W)	(m)		(°)	(°)	(°)			(°)	(°)	(°)	(°)
Santa Maria rhyolites subtype													
XG-01	29.273	52.649	476	9/9	164.2	45.6	1.8	851	R	−75.9	42.8	2.2	1.4
XG-02	29.274	52.650	475	9/9	167.1	43.1	2.7	375	R	−77.8	54.3	3.3	2.1
XG-03	29.273	52.649		12/9	168.3	49.7	3.5	213	R	−79.8	27.5	4.7	3.1
XG-04	29.272	52.648	449	11/11	162.0	46.0	2.6	310	R	−74	39.9	3.3	2.1
XG-05	29.271	52.647	438	10/9	163.7	37.6	2.1	577	R	−73.1	62.6	2.5	1.5
XG-06	29.277	52.652		12/9	162.2	43.1	2.3	482	R	−73.6	48.1	2.9	1.8
XG-07	29.275	52.651	488	26/9	158.0	41.1	2.5	424	R	−69.5	48.4	3	1.9
XG-08	29.282	52.641		13/9	167.8	43.1	4.8	118	R	−78.4	55.7	5.9	3.7
XG-09	29.279	52.642	445	12/9	185.3	43.5	3.2	257	R	−83.9	178.9	4	2.5
XG-10	29.290	52.644	522	11/11	168.3	47.0	2.7	286	R	−79.7	40.7	3.5	2.3
XG-15	29.246	52.660	456	9/9	180.0	55.4	3.1	270	R	−83.3	307.5	4.5	3.2
XG-16	29.246	52.660	465	9/9	174.0	55.0	1.6	1066	R	−81.9	344.5	2.2	1.6
XG-17	29.246	52.660	465	9/9	175.8	55.6	2.4	447	R	−82.3	333.2	3.5	2.5
XG-18	29.246	52.660	466	9/9	169.9	54.1	3.1	272	R	−79.9	2.6	4.4	3.1
Caxias do Sul dacites subtype													
XG-12	29.263	52.639	366	20/11	0.5	12.0	4.1	125	I	54.7	308.3	4.2	2.1
XG-13	29.263	52.639	364	13/11	342.0	22.4	5.1	83	I	45.6	281.8	5.4	2.8
XG-14	29.263	52.639	355	13/9	343.5	8.7	4.8	117	I	52.8	279.5	4.8	2.4
XG-19	29.262	52.639	364	10/9	350.2	−6.5	3.7	194	I	−62.4	105.8	3.7	1.9
XG-20	29.267	52.642	388	10/10	4.9	−1.4	4.3	129	I	−61.0	137.6	4.3	2.1
XG-21	29.267	52.642	392	9/9	1.5	8.3	4.7	120	I	56.5	310.1	4.8	2.4
XG-22	29.267	52.642	395	9/9	4.3	−10.4	5.1	103	I	−65.6	137.9	5.2	2.6
XG-23	29.267	52.642	394	9/8	5.6	3.1	8.4	44	I	58.7	318.2	8.4	4.2
XG-24	29.267	52.642	403	10/10	0.7	−8.4	5.3	83	I	−65.0	129.0	5.4	2.7
XG-29	29.259	52.634	370	11/9	355.2	12.7	5.8	80	I	54.0	299.3	5.9	3
XG-30	29.261	52.628	370	11/10	313.6	35.2	6.8	51	I	23.9	259.0	7.9	4.5
XG-31	29.264	52.624		11/9	353.0	10.2	11.6	28	I	54.9	295.1	11.8	6
XG-32	29.264	52.610	403	11/11	336.2	37.5	8.7	29	I	34.8	280.1	10.2	6
XG-33	29.264	52.582	467	11/11	326.9	13.7	18.3	7	I	41.8	260.8	18.7	9.5
Barros Cassal andesite subtype													
XG-11	29.250	52.643	429	11/11	176.6	47.5	2.9	254	R	−86.9	47.9	3.7	2.4
XG-34	29.124	52.562	473	15/9	1.9	−46.9	1.9	724	N	−88.1	185.4	2.5	1.6

Table 1. Lat. is the latitude and Long. is the longitude; El. is altitude; N, number of specimens analyzed; n, number of specimens for mean calculations; Dec. and Inc., Declination and Inclination; α_{95} and k, Fisher's (1953) statistical parameters; Pol., polarity (N = normal, I = intermediate and R = reverse); dm and dp, confidence oval of 95% for PGVs..

UK) and a DT-2000 tumbler (ASC Scientific) demagnetizers. Another specimen was submitted to a detailed (pilot) stepwise thermal demagnetized from 100 °C up to 600° or 700 °C in step of 50 °C up to 500 °C and 25° or 30 °C up to 600 or 700 °C with a TD-48 oven (ASC Scientific). The remanences were measured in JR5A or JR6A magnetometers (Agico Czech Republic). After this procedure at least 8–9 specimens from each site were subjected to a.f. or thermal demagnetization in longer steps (as a semi-pilot).

By visual inspection of orthogonal demagnetization diagrams (Zijderveld, 1967) the vectorial behavior of the NRM was determined, and the characteristic remanent magnetization (ChRM) for each specimen was determined using principal component analysis (Kirschvink et al., 1980). The site mean directions were calculated assuming that the distribution of directions is well described by a Fisherian distribution (Fisher, 1953). In both cases, the routine was provided by home-made software.

4. Results

4.1. Rock magnetism properties

Examples of k-T curves for basalts, rhyolites and basaltic autobreccia outcropping in area #II are shown in Fig. 3.

Samples show a well-defined peak around −150 °C (Fig. 3A) which indicates the Verwey transition, characteristic of almost pure magnetite. The high k-T curves show a small Hopkinson peak, and all of them display a decrease in the intensity of susceptibility around 550°C–580 °C (Fig. 3A) indicating the presence of Ti-poor titanomagnetite or magnetite. The cooling and heating curves are nearly reversible in almost all specimens. The basaltic autobreccias (Fig. 3B) show a small Verwey transition, however, there is an increase in the intensity of susceptibility up to ~300 °C, indicating of creating a new magnetic phase during the heating. The intensity of susceptibility decreases ~580 °C which is the Curie temperature for magnetite. The magnetic mineral of the dacites from Caxias do Sul is similar to the basalts and the rhyolites (Guimarães et al., 2018b).

The obtained pARM curves are shown in Fig. 4A, C, and E for basalts, rhyolites, and basaltic autobreccias, respectively. These curves show that the majority of samples are characterized by low coercivity, indicating that magnetite grains are relatively large (around 2–5 m, μJackson et al., 1989).

The IRM patterns are shown in Fig. 4B, D and F for basalts, rhyolites, and basaltic autobreccias, respectively. For the basalts (Fig. 4B) more than 95% of the magnetization reaches the saturation (SIRM) in fields <150 mT. For the rhyolites (Fig. 4D) saturation is reached in fields <200–250 mT, however, the IRM curves of the basaltic autobreccias (Fig. 4F) do not reach the saturation even in fields as high as 1.6 T, showing the presence of high coercive minerals such as hematite. The IRM curves indicate the presence of saturated coarse and fine magnetite grains, which is in agreement with the pARM spectra (Fig. 4A, D) except for the basaltic autobreccias, in which hematite occurs in low contents since it was not observed in the kxT curve (Fig. 3B). The hematite only

Table 2
Paleomagnetic results for the volcanic rocks from area #II.

Site	Lat.	Long.	El.	N/n	Mean magnetization direction					Virtual Palemagnetic Pole			
					Dec.	Inc.	α_{95}	k	Pol	PLat.	Plong.	dm	dp
	(°S)	(°W)	(m)		(°)	(°)	(°)			(°)	(°)	(°)	(°)
Basalt Torres Formation													
TP-01	29.727	52.414	157	13/10	353.3	−47.8	3.1	249	N	−84.1	44.0	4	2.6
TP-02	29.606	52.644	163	15/11	330.8	−44.1	4.1	123	N	−64.0	38.4	5.2	3.3
TP-03	29.554	52.646	208	14/11	1.9	−50.0	5.8	64	N	−87.9	254.4	7.7	5.2
TP-04	29.549	52.649	274	23/10	5.9	−62.8	4.7	107	N	−74.6	291.2	7.4	5.8
TP-05	29.547	52.656	248	12/11	356.3	−44.3	3.0	234	N	−85.2	83.7	3.8	2.4
TP-07	29.553	52.635	158	13/13	12.7	−37.7	4.0	107	N	−75.8	184.0	4.7	2.8
TP-08	29.550	52.636	177	13/10	346.7	−39.4	4.7	105	N	−76.0	65.5	5.7	3.4
TP-09	29.543	52.639	232	13/12	356.1	−36.8	4.4	100	N	−80.3	105.2	5.1	3.0
TP-10	29.542	52.638	256	14/13	359.6	−39.2	2.8	220	N	−82.6	124.5	3.3	2.0
TP-11	29.532	52.637	282	14/11	353.4	−44.3	2.5	344	N	−83.2	66.7	3.1	1.9
TP-12	29.525	52.640	319	14/14	354.4	−60.1	3.1	169	N	−77.6	327.3	4.6	3.5
TP-13	29.522	52.641	318	10/10	182.7	43.8	2.2	505	R	−85.4	159.8	2.7	1.7
Basalt Vale do Sol Formation													
XG-26	29.521	52.644	375	13/12	354.4	−50.9	3.6	146	N	−84.7	12.5	4.9	3.3
XG-27	29.522	52.651	432	15/13	354.7	−55.1	3.1	182	N	−82.4	341.7	4.4	3.1
XG-28	29.519	52.651	456	10/9	166.7	40.4	1.9	699	R	−76.4	62.7	2.4	1.4
Basaltic Autobreccias													
TP-14	29.520	52.651	431	16/10	327.7	−24.6	7.0	49	N	−55.8	59.5	7.5	4.0
TP-15	29.516	52.651	465	20/10	174.9	41.4	4.5	117	R	−82.7	87.6	5.5	3.3
TP-16	29.513	52.650	489	10/10	181.5	42.9	3.6	180	R	−85.2	143.7	4.5	2.8
Santa Maria rhyolites subtype													
TP-06	29.527	52.658	616	13/12	167.9	51.8	3.7	140	R	−79.2	18.6	5.0	3.4
TP-17	29.510	52.649	509	14/11	176.8	44.2	4.1	123	R	−85.4	88.2	5.2	3.3
TP-18	28.825	52.679	586	13/12	168.1	64.7	2.8	246	R	−69.9	331.7	4.5	3.6

Lat. is the latitude and Long. is the longitude; El. is altitude; N, number of specimens analyzed; n, number of specimens for mean calculations; Dec. and Inc., Declination and Inclination; α_{95} and k, Fisher's (1953) statistical parameters; Pol., polarity (N = normal, R = reversed); dm and dp, confidence oval of 95% of the PGVs.

appears in the breccias and its origin probably is secondary origin related to the brecciation process.

Typical hysteresis curves from basalts, rhyolites, and basaltic autobreccias are illustrated in Fig. 5A, B, C and D, respectively. For all samples the shape of the curves reveals that ferromagnetic grains carry bulk magnetic susceptibility, with contribution of paramagnetic minerals up to 5% (Fig. 5B), except for TP-15 (Fig. 5D) that does not reach saturation. In general, the curves are relatively narrow-waisted, typical of low-coercivity ferromagnetic minerals, which is in accordance with the other experiments such as the IRM curves (Fig. 4B and D). Except for the basaltic autobreccias whose shape of the hysteresis (Fig. 5D) is typical of hematite, in which a field of 1T (maximum VSM field) was not sufficient to saturate the sample as also indicated by IRM curves (Fig. 4F).

The hysteresis' parameters: Mrs (remanent saturation magnetization), Ms (saturation magnetization), Hc (coercitive force) were calculated after correction for the paramagnetic contribution, and Hcr (remanent coercivity) was determined by applying progressively increasing backfield after saturation. These parameters were plotted in the Dunlop's (2002) diagram to verify the magnetic domain structures of the magnetite (Fig. 5E). This plot shows that all magnetite grains fall into the pseudo-single-domain (PSD) field.

FORC diagrams (Fig. 5E) show dispersion on the interaction (Bu) and coercivity (Bc) axes, in an approximately triangular geometry, typical of PSD (titano)magnetites. Therefore, the domain structure of the (titano) magnetites given by the FORCs is consistent with the information given by the hysteresis parameters in the Day Diagram (Fig. 5E). Even though, Roberts et al. (2018) pointed out that Day's plot is ambiguous what do not happen in our case. The PSD magnetic grain is known for its extreme magnetic stability (e.g. Nagy et al., 2017), ensuring the reliability of the paleomagnetic data presented here.

The same experiments were previously performed in samples of the silicic sequences from area #I (for results and discussions the reader is referred to Guimarães et al., 2018b). The results obtained for area #I are similar to those obtained at area #I.

4.2. Paleomagnetism

Paleomagnetic results for the volcanic rocks from area #I and area #II (Figs. 1 and 2) are presented in Tables 1 and 2, respectively. Representative examples of the NRM behavior in both a.f. and thermal demagnetization are shown in Fig. 6. In general, both demagnetization processes are efficient and similar for rocks from area #I, however, the thermal demagnetization was particularly most efficient in demagnetizing rocks from area #II (Fig. 6).

The characteristic remanent magnetization (ChRM) is isolated after removing soft components yielded by low temperature (Fig. 6A, B and D) or low coercivity (Fig. 6C). The demagnetization trajectories to the origin allow recognizing that the ChRM (using anchored lines) is quite well precise, with low values (<5°) of maximum angular deviation (MAD). Specimens with MAD>10° and low magnetic stability were discarded. In general, samples are near univectorial behavior, median destructive fields range between about 40 and 60 mT (Fig. 6C), except for autobreccias which show three vectors (Fig. 6D). The ChRM is obtained mostly between around 20 and 100 mT, and unblocking temperatures are spread in the range of 500 °C and 625° - 630 °C (Fig. 6A, B, and D). This pattern suggests that oxidized low-Ti magnetite is the carriers of the remanence, which is also shown by the rock-magnetism measurements.

The samples exhibit only one stable component of magnetization ~ N–S oriented, after elimination of a soft viscous magnetization at fields up to 15–20 mT. The ChRM of the autobreccias (TP-16, Fig. 6D) is mainly carried by magnetite since it has similar mean direction of the other rocks (Figs. 6 and 7). The ChRM is interpreted as a thermo-

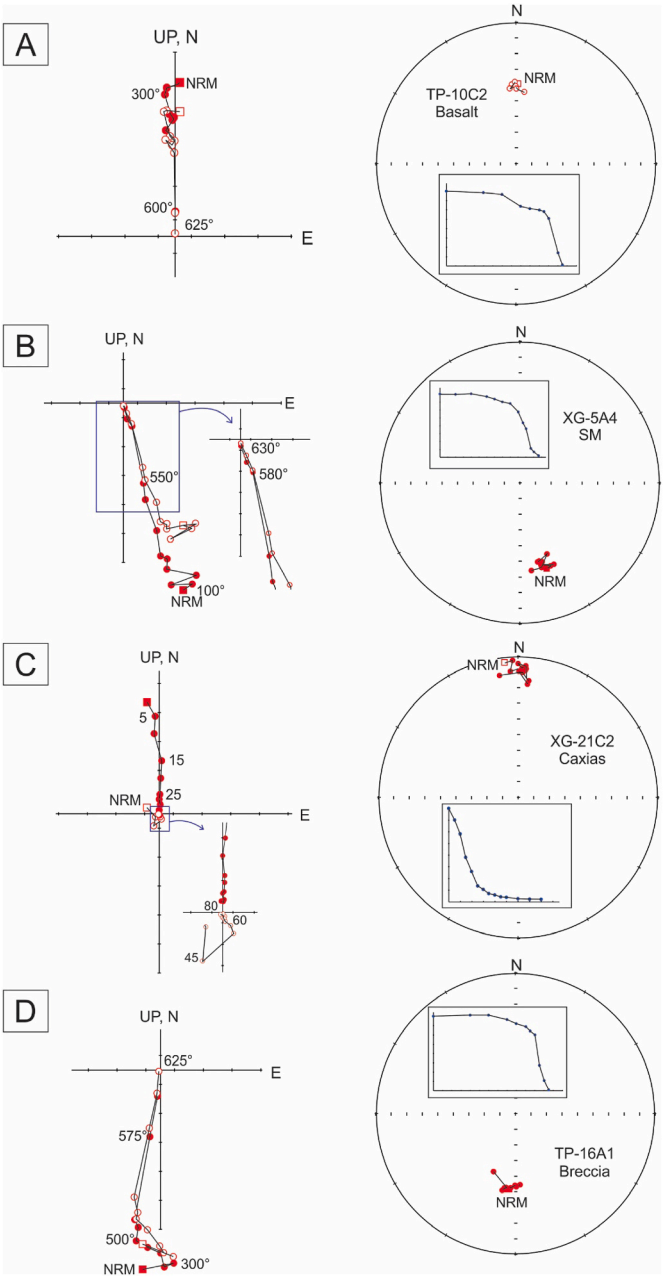


Fig. 6. Representative examples of NRM behavior. Orthogonal diagrams for the volcanic rocks and stereographic projections (low hemisphere) of the magnetization vector during demagnetization. The variation of the magnetization intensity as function of the demagnetization steps is shown in the inset of stereograms. Open (full) symbols correspond to negative (positive) inclinations. Magnetization units are in A/m, and field in mT. ***A-*** Basalts, Vale do Sol Fm; ***B-*** Santa Maria (SM) Rhyolites; ***C-*** Caxias do Sul Dacites; ***D-*** Basaltic autobreccias.

remanent magnetization acquired during the cooling of the flows.

The magnetization of each specimen yields well-defined site means (Fig. 7, with $\alpha_{95} \ll 10^{\circ}$ and high k, Tables 1 and 2). The ChRM in both studied areas is of dual polarity reverse (positive inclination) and normal (negative inclination) for Torres and Vale do Sol basalts, Barros Cassal andesites and Santa Maria rhyolites (Fig. 7A and B) while all analyzed site from Caxias do Sul dacites (that outcrop only in area #I) show "anomalous" low-inclination directions which are also found in other regions of the Paraná basin (e.g. Ernesto et al., 1990). This direction can be related to a transitional direction that represents an excursion of the geomagnetic field at that time. The reverse and normal polarities show

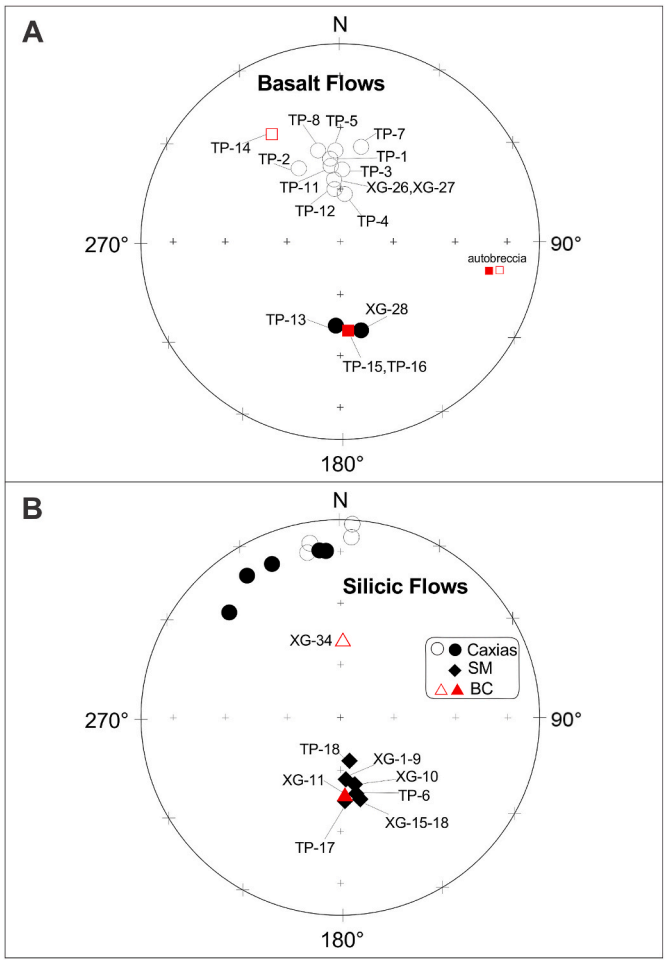


Fig. 7. Mean magnetization directions of the sampled sites. Open (full) symbols correspond to negative (positive) inclinations. ***A-*** Basalts and ***B-*** Silicic rocks. Some sites were grouped to clarify the figure. Caxias = Caxias do Sul unit, SM= Santa Maria unit, BC=Barros Cassal unit. Stereographic projections in low hemisphere.

positive reversal test, which means that they cannot be distinguishable at 95% confident level.

5. Discussion

Rock magnetism investigations have shown that relatively coarse (2–5 μm, PSD domain) Ti-poor titanomagnetite grains are the main magnetic carriers in the volcanic rocks (Fig. 4), and they contribute to more than 98% of the magnetic susceptibility (Fig. 5). These grains are the carriers of the thermal remanent magnetization, however, hematite can be found in basaltic autobreccias (Figs. 4 and 5), which is probably secondary in origin.

Paleomagnetic studies show a soft component given by low temperature (Fig. 6A, B and D) or low coercivity (Fig. 6C), which we interpret as a viscous remanence. The ChRM of the basalts, Barros Cassal andesites, and Santa Maria rhyolites is oriented around N–S with different inclinations and polarities (Figs. 6 and 7). The difference in the inclinations is probably due to secular variation of the geomagnetic field at that time. The ChRM in the analyzed rocks is primary in origin and is thermoremanent (TRM) since magnetizations of both normal and reverse polarity are registered (Figs. 6 and 7), and it is stable in progressive demagnetization, with medium MDFs (Fig. 6, Fuller et al., 2002).

The majority of basaltic rocks have normal polarity (negative inclination, Fig. 6A, 7A and 2), however, 2 sites (TP-13 and XG-28) show

reverse polarity (positive inclination, Fig. 7A). Between the sites TP-13 (reverse) and XG-26 (normal) there is an interbedded sedimentary rock indicating that both lava flows were emplaced at different time. After the emplacement of the XG-26 and XG-27 (normal, Figs. 7A and 2) the geomagnetic field reversed again and the XG-28 was emplaced. In addition, the directions of ChRM of TP-13 and XG-28 (Fig. 7A) are different at 95% confidence level as indicated by the F-test of McFadden and Lowes (1981).

The Caxias do Sul dacites show an "anomalous" ChRM direction which we interpret as an excursion of the geomagnetic field (Fig. 7B). The ChRMs for the two sites of the Barros Cassal andesites (area #I) show normal (XG-34), and reverse polarities (XG-11, Fig. 7B). Even though the ChRM of XG-34 is near to the ChRM of the basalts with normal polarity (XG-26 and XG-27; Fig. 7A), they are different of the basalts at 95% confidence level as indicated by the F-test of McFadden and Lowes (1981). On the other hand, the ChRMs of Barros Cassal andesites XG-11 (Fig. 7B) and Torres basalt TP-13 (Fig. 7A, both reverse) cannot be distinguishable at 95% confidence level (Fig. 7).

The ChRM of all sites from Santa Maria rhyolites, in both areas, show reverse polarity (Fig. 7B). The basaltic autobreccias recorded the same polarity as their respective flows.

Fig. 2 shows the profiles coded as letters indicated in Fig. 1, with the obtained polarities for all studied sequences. Based on the polarities of the geomagnetic field associated with the stratigraphy, it can be inferred that Torres Formation (profile D, Fig. 2) was emplaced in two distinct events. The first lavas (with pahoehoe morphology) were emplaced in normal polarity (Fig. 7A and 2; TP-01 to TP-05, and TP-07 to TP-12). Then an inversion occurred and the polarity became reverse (Fig. 7A, TP-13). After that, during the emplacement of the rubbly lavas from the Vale do Sol Formation, the polarity changed again becoming normal, (Fig. 7A and 2, profile D, XG-26, XG-27 and TP-14). At last, another geomagnetic field inversion occurred and the last basalt flows were emplaced under reverse polarity (XG-28, TP-15 and TP-16). The Santa Maria rhyolites (profile D, Fig. 2, TP-06, TP- 16 and TP-17) were emplaced directly over the basalt with reverse polarity which suggests that the sources of basaltic (Vale do Sol Formation) and rhyolitic (Santa Maria) flows may be simultaneously active.

In the profile C and the low part of profile B (Fig. 2), the Caxias do Sul dacites recorded an excursion of the geomagnetic field that was not registered by other rocks even in the same area, suggesting that just the source of the Caxias do Sul dacites was active at that time. After that, the polarity became reverse and Barros Cassal andesites (profile B, Fig. 2, XG-11) and Santa Maria rhyolites (XG-01 to XG-10, XG-15 to XG-18) were emplaced, suggesting the contemporaneity between the sources of these rocks. The same polarity was registered by the basalts TP-13 (Torres Formation, Fig. 7A) suggesting that basaltic source could be active simultaneously with the Barros Cassal andesites and Santa Maria rhyolites sources.

In the profile A (Figs. 2 and 7B), before the emplacement of Santa Maria rhyolites, just one pulse of the Barros Cassal andesites occurred (Figs. 2 and 7A), since it was not recorded in the other profiles. The polarity of the geomagnetic field was normal (XG-34) but statistically different from the basalts. Afterwards, the polarity became reverse and the rhyolite Santa Maria was emplaced covering the whole profiles with the same polarity.

It is noteworthy that the time span of the basaltic and silicic volcanism was very rapidly, since the ChRM of the rocks is N–S oriented and there is little variation in the inclination. Similar results were obtained elsewhere in the Province (e.g Ernesto et al., 1990), corroborating with the rapid emplacement of the PMP volcanism as also show by radiometric ages (Rocha et al., 2020).

6. Final remarks

The high-quality of the ChRM results from the studied region allows us to interpret that the emplacement of all volcanic rocks occurred very rapidly, since the ChRM is N–S oriented with little variation in inclination. The sources of basaltic Torres, Vale do Sol Formations and Barros Cassal andesites were active in two distinct events as shown by the polarity changes of the geomagnetic field. The passage of Torres (pahoehoe) to Vale do Sol Formation (rubbly) was marked by a change of polarity of the geomagnetic field. As Santa Maria rhyolites were emplaced directly over the Vale so Sol basalts with the same reverse polarity we can infer that their sources may be active contemporaneously or simultaneously. The Caxias do Sul dacites recorded an excursion of the geomagnetic field. The feeder systems of Vale do Sol basalts, Barros Cassal andesites and Santa Maria rhyolites were probably active contemporaneously or simultaneously. The Santa Maria rhyolites were emplaced and cover the whole studied area with the same polarity.

CRediT authorship contribution statement

M. Irene B. Raposo: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Thales Pescarini:** Writing – original draft, Methodology, Formal analysis. **Letícia F. Guimarães:** Writing – review & editing, Writing – original draft, Visualization. **Melina C.B. Esteves:** Writing – review & editing, Writing – original draft, Visualization, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This research benefits from funding by São Paulo Research Foundation (FAPESP, grants 2012/06082–6 and a scholarship given to Pescarini, T., 2017/01227–0). We are grateful to Valdecir A. Janasi, Odirney B. da Silva and Edgardo Cañón Tapia for assistance during fieldwork. We thanks to Silvana Geuna and an anonymous reviewer whose comments improved the manuscript.

References

Baksi, A., 2018. Paraná flood basalt volcanism primarily limited to ~1Myr beginning at 135 Ma: new 40Ar/39Ar ages for rocks from Rio Grande do Sul, and critical evaluation of published radiometric data. J. Volcanol. Geoth. Res. 355, 66–77.

Barreto, C.J.S., Lima, E.F., Scherer, C.M., Rossetti, L.M.M., 2014. Lithofacies analysis of basic lava flows of the Paraná igneous province in the south hinge of Torres Syncline, Southern Brazil. J. Volcanol. Geoth. Res. 285, 81–99. https://doi.org/10.1016/j.jvolgeores.2014.08.008.

Barreto, C.J.S., Lafon, J.M., Lima, E.F., Sommer, C.A., 2016. Geochemical and Sr-Nd-Pb isotopic insight into the low-Ti basalts from southern Paraná Igneous Province, Brazil: the role of crustal contamination. Int. Geol. Rev. 58, 1324–1349. https://doi.org/10.1080/00206814.2016.1147988.

Bellieni, G., Comin-Chiaramonyi, P., Marques, L.S., Melfi, A.J., Nardy, A.J.R., Papatrechas, C., Piccirillo, E.M., Roisenberg, A., Stolfa, D., 1984. Petrogenetic aspects of acid and basaltic lavas from the Paraná plateau (Brazü): geological, mineralogical and petrochemical relationships. J. Petrol. 27, 915–944.

Bryan, S.E., Peate, I.U., Peate, D.W., Self, S., Jerram, D.A., Mawby, M.R., Marsh, J.S.G., Miller, J.A., 2010. The largest volcanic eruptions on Earth. Earth Sci. Rev. 102, 207–229. https://doi.org/10.1016/j.earscirev.2010.07.001.

Dunlop, D.J., 2002. Theory and application of the Day plot (Mrs/Ms versus Hcr/Hc) 1. Theoreticalcurves and tests using titanomagnetite data. J. Geophys. Res. 107 https://doi.org/10.1029/2001JB000486. EPM4–1–EPM4–22).

Ernesto, M., Pacca, I.G., 1988. Paleomagnetism of the Parana basin flood volcanism, southern Brazil. In: Piccirillo, E., Melfi, A.J. (Eds.), The Mesozoic Flood Volcanism of the Parana Basin. Universidade de São Paulo, pp. 229–255.

Ernesto, M., Pacca, I.G., Hiodo, F.Y., Nardy, A.J.R., 1990. Palaeomagnetism of the Mesozoic Serra Geral formation, southern Brazil. Phys. Earth Planet. In. 64, 153–175. https://doi.org/10.1016/0031-9201 (90) 90035-V.

Ewart, A., Milner, S.C., Armstrong, R.A., Duncan, A.R., 1998. Etendeka volcanism of the Goboboseb Mountains and Messum igneous complex, Namibia. Part II: voluminous quartz latite volcanism of the Awahab magma system. J. Petrol. 39, 227–253. https://doi.org/10.1093/petroj/39.2.227.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. Roy. Soc. Lond. Ser. A 217, 295–305.

Frank, H.T., Gomes, M.E.B., Formoso, M.L.L., 2009. Review of the areal extent and the volume of the Serra Geral formation, Paraná basin, south America. Pesqui. em Geociencias 36, 49–57.

Fuller, M., Kidane, T., Ali, J., 2002. AF demagnetization characteristics of NRM, compared with anhysteretic and saturation isothermal remanence: an aid in the interpretation of NRM, Physics and Chemistry of the Earth. Parts A/B/C 27, 1169–1177.

Garland, F., Hawkesworth, C.J., Mantovani, M.S.M., 1995. Description and petrogenesis of the paraná rhyolites, southern Brazil. J. Petrol. 36, 1193–1227.

Giordano, D., Vona, A., Gonzalez-Garcia, D., Allabar, A., Kolzenburg, S., Polo, L.A., Janasi, V.A., Behrens, H., De Campos, C.P., De Cristofaro, S., Guimarães, L.F., Nowak, M., Müller, D., Günther, A., Massotta, M., Roverato, M., Romano, C., Dingwell, D.B., 2021. Viscosity of Palmas-type magmas of the paraná magmatic province (Rio Grande do Sul state, Brazil): Implications for high-temperature silicic volcanism. Chem. Geol. 560, 119981 https://doi.org/10.1016/j.chemgeo.2020.119981.

Guimarães, L.F., 2014. Características físicas e químicas e modelo eruptivo para os riolitos tipo Santa Maria (Província Magmática Paraná) na região de Gramado Xavier. RS. Thesis Universidade de São Paulo.

Guimarães, L.F., De Campos, C.P., Janasi, V.A., Lima, E.F., Dingwell, D.B., 2018a. Flow and fragmentation patterns in the silicic feeder system and related deposits in the Paraná-Etendeka Magmatic Province, São Marcos, South Brazil. J. Volcanol. Geoth. Res. 358 https://doi.org/10.1016/j.jvolgeores.2018.03.021.

Guimarães, L.F., Raposo, M.I.B., Janasi, V.A., Cañón-Tapia, E., Polo, L.A., 2018b. An AMS study of different silicic units from the southern Paraná-Etendeka Magmatic Province in Brazil: Implications for the identification of flow directions and local sources. J. Volcanol. Geoth. Res. 355, 304–318. https://doi.org/10.1016/j.jvolgeores.2017.11.014.

Guimarães, L.F., Hornby, A., Kueppers, U., Alves, A., Janasi, V.A., Dingwell, D.B., 2019. Generation of block-and-ash flows at the onset of silicic volcanism in the Paraná Magmatic Province (Brazil): evidence from photoanalysis of Caxias do Sul breccias. Bull. Volcanol. 81, 65. https://doi.org/10.1007/s00445-019-1332-7.

Harrison, R.J., Feinberg, J.M., 2008. FORCinel: an improved algorithm for calculating first-order reversal curve distributions using locallyweighted regression smoothing. Geochem. Geophys. Geosyst. 9, Q05016 https://doi.org/10.1029/2008GC001987.

Hrouda, F., 1994. A technique for the measurement of thermal changes of magnetic susceptibility of Weakly magnetic rocks by the CS-2 apparatus and KLY-2 Kappabridge. Geophys. J. Int. 118, 604–612. https://doi.org/10.1111/j.1365-246X.1994.tb03987.x.

Jackson, M., Sprowl, D., Ellwood, B.B., 1989. Anisotropy of partial anhysteretic remanence and susceptibility in compact black shales: grain-size and composition-dependent magnetic fabric. Geophys. Res. Lett. 16, 1063–1066.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophys. J. Roy. Astron. Soc. 62, 699–718.

Lima, E.F., Philipp, R.P., Rizzon, G.C., Waichel, B.L., Rossetti, L.M.M., 2012. Sucessões vulcânicas, modelo de alimentação e geração de domos de lava ácidos da Formação Serra Geral na região de São Marcos-Antônio Prado (RS). Geol. Usp. Série Científica 12, 49–64. https://doi.org/10.5327/Z1519-874X2012000200004.

Luchetti, A.C.F., Gravley, D.M., Gualda, G.A.R., Nardy, A.J.R., 2018a. Textural evidence for high-grade ignimbrites formed by low-explosivity eruptions, Paraná Magmatic Province, southern Brazil. J. Volcanol. Geoth. Res. 355, 87–97. https://doi.org/10.1016/J.JVOLGEORES.2017.04.012.

Luchetti, A.C.F., Nardy, A.J.R., Madeira, J., 2018b. Silicic, high- to extremely high-grade ignimbrites and associated deposits from the Paraná Magmatic Province, southern Brazil. J. Volcanol. Geoth. Res. 355, 270–286. https://doi.org/10.1016/J.JVOLGEORES.2017.11.010.

Mantovani, M.S.M., Marques, L.S., De Sousa, M.A., Civetta, L., Atalla, L., Innocenti, F., 1985. Trace element and Strontium isotope constraints on the origin and evolution of paraná continental flood basalts of Santa Catarina state (southern Brazil). J. Petrol. 26, 187–209. https://doi.org/10.1093/petrology/26.1.187.

McFadden, P.L., Lowes, F.J., 1981. The discrimination of mean directions drawn ITom Fisher distributions. Geophys. J. Roy. Astron. Soc. 67, 19–33.

Nagy, L., Williams, W., Muxworthy, A.R., Fabian, K., Almeida, T.P., Ó Conbhui, P., Shcherbakov, V.P., 2017. Stability of equidimensional pseudo-single-domain magnetite over billion-year timescales. Proc. Natl. Acad. Sci. U. S. A 114, 10,356–10,360.

Nardy, A., Machado, F., Oliveira, M., 2008. As rochas vulcânicas mesozóicas ácidas da Bacia do Paraná: litoestratigrafia e considerações geoquímico-estratigráficas. Rev. Bras.Geosci. 38, 178–195.

ocha-Júnior, E.R.V., Puchtel, I.S., Marques, L.S., Walker, R.J., Machado, F.B., Nardy, A.J.R., Babinski, M., Figueiredo, A.M.G., 2012. Re–Os isotope and highly siderophile element systematics of the Paraná continental flood basalts (Brazil). Earth Planet Sci. Lett. 337–338, 164–173. https://doi.org/10.1016/J.EPSL.2012.04.050.

Peate, D., Hawkesworth, C., Mantovani, M., Shukowsky, W., 1990. Mantle plumes and flood-basalt stratigraphy in the Paraná, South America. Geology 18, 1223–1226. https://doi.org/10.1130/0091-7613(1990)018b1223. MPAFBSN2.3.CO;2).

Peate, D.W., Hawkesworth, C.J., Mantovani, M.S.M., 1992. Chemical stratigraphy of the Paran'a lavas (South America): classification of magma types and their spatial distribution. Bull. Volcanol. 55, 119–139.

Pescarini, T., Raposo, M.I.B., 2017. Estudos magnéticos da Formação Serra Geral (Bacia do Paraná) na Região de Santa Cruz do Sul-Herveira, RS, Brasil: resultados Preliminares. Latinmag. Lett. 7 (2017), 1–5 (Proceedings Juriquilla, Qro, Mexico).

Piccirillo, E.M., Melfi, A.J., 1988. The Mesozoic Flood Volcanism of the Paran'a Basin: Petrogenetic and Geophysical Aspects. Instituto Astronˆomico e Geofísico – Universidade de São Paulo, 600p., São Paulo.

Polo, L.A., Janasi, V. de A., 2014. Volcanic stratigraphy of intermediate to acidic rocks in southern Paraná Magmatic Province, Brazil. Geol. Usp. Série Científica 14, 83–100. https://doi.org/10.5327/Z1519-874X201400020005.

Polo, L.A., Janasi, V.A., Giordano, D., Lima, E.F., Cañón-Tapia, E., Roverato, M., 2018. Effusive silicic volcanism in the Paraná Magmatic Province, South Brazil: evidence for locally-fed lava flows and domes from detailed field work. J. Volcanol. Geoth. Res. 355, 204–218. https://doi.org/10.1016/J.JVOLGEORES.2017.08.007.

Renne, P.R., Ernesto, M., Pacca, I.G., Coe, R.S., Glen, J.M., Prévot, M., Perrin, M., 1992. The age of paraná flood volcanism, rifting of Gondwanaland, and the Jurassic-Cretaceous boundary. Science 258, 975–979.

Roberts, A.P., Tauxe, L., Heslop, D., Zhao, X., Jiang, Z.X., 2018. A critical appraisal of the"Day"diagram. J. Geophys. Res. Solid Earth 123, 2618–2644.

Rocha, B.C., Davies, J.H.F.L., Janasi, V.A., Schaltegger, U., Nardy, A.J.R., Greber, N.D., Lucchetti, A.C.F., Pólo, L.A., 2020. Rapid eruption of silicic Valanginian event. Geology 48, 1174–1178. https://doi.org/10.1130/G47766.1.

Rossetti, L.M.M., Lima, E.F., Waichel, B.L., Hole, M.J., Simões, M.S., Scherer, C.M.S., 2018. Lithostratigraphy and volcanology of Serra Geral group, paraná-Etendeka igneous province in southern Brazil: towards a formal stratigraphical framework. J. Volcanol. Geoth. Res. 355, 98–114.

Simões, M.S., Lima, E.F., Sommer, C.A., Rossetti, L.M.M., 2018. Structures and lithofacies of inferred silicic conduits in the Paraná-Etendeka LIP, southernmost Brazil. J. Volcanol. Geoth. Res. 355, 319–336. https://doi.org/10.1016/J.JVOLGEORES.2017.12.013.

Zijderveld, J.D.A., 1967. AC demagnetization of rocks: analysis of results. In: Collinson, D.W., Creer, K.M., Runcorn, S.K. (Eds.), Methods in Paleomagnetism. Elsevier, Amsterdam, pp. 254–286.